MA491: Research Seminar

AY27-1 Course Syllabus

LTC Dusty Turner, PhD

1 Course Information

Course	MA491: Research Seminar
Term	AY27-1 (Fall 2027)
Credits	3.0
Meeting Time	A2 hour
Location	TH239B

2 Instructor

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3 Course Description

MA491 is a capstone research course for non-thesis cadets in the Department of Mathematical Sciences. Cadets stand up a working computational environment from bare hardware (Raspberry Pi), then run the research cycle four times with the scaffolding coming off a step at a time: guided, semi-independent, independent, and finally a capstone that culminates in a public Demo Day. Capstone scoping starts at the midpoint of the semester, with opportunities to work that project for a sponsor. Cadets pursuing applied or theoretical mathematics topics may shape their independent and capstone work accordingly, in coordination with the instructor.

Cadets use AI tools throughout the course as collaborators in the research and analytical workflow. The focus is on framing real questions, working with real problems, and producing real findings, not on learning syntax. By the end of the semester, every cadet will have a portfolio of hosted projects, a personal website, and experience presenting findings to real stakeholders.

4 Course Objectives

Upon completion of this course, cadets will be able to:

1. **Conduct end-to-end investigations:** frame a question, gather evidence or data, analyze, and communicate findings.
2. **Apply quantitative and modeling judgment:** choose appropriate methods, evaluate fit, and reason about uncertainty.
3. **Use AI tools as professional collaborators in the research workflow.**
4. **Work with real-world problems and data sources:** files, APIs, databases, and open literature.
5. **Produce shareable scholarly and analytical artifacts:** dashboards, web apps, papers, and reproducible reports that communicate results to technical and non-technical audiences.

6. **Conduct an independent or team-based research project:** scoping, executing, and presenting findings publicly.
7. **Operate with a professional research workflow:** version control, deployment, and reproducibility.
8. **Reason about where code actually runs.** Recognize that memory, processing speed, storage, algorithmic and rendering cost, concurrent users, and hosting limits are real constraints on deployed work, and treat them as standard considerations when something breaks or hangs.
9. **Maintain a portfolio that demonstrates research and analytical capability** to advisors, future employers, and program partners.

5 Course Arc

Each block below covers a body of work and ends with something built, hosted, and briefed. We move at the pace of the class, spending more time where a topic warrants it and following worthwhile questions as they come up in the natural progression of the course.

Each class day is split. The first part is conceptual work: what is happening under the hood, demonstrations, and discussion. The second part is hands-on reps on the current block's project.

5.1 Block 1: Setup & Foundations

Standing up a working environment from bare hardware, learning the tools the rest of the course runs on, and building the site everything you produce will be published to.

- Unboxing the Raspberry Pi and what a computer actually is
- Linux, SSH, and installing RStudio
- Git and GitHub
- Getting data: databases and APIs
- Data exploration and dashboards
- Hosting: local, Tailscale, and GitHub Pages
- What your machine and your host can and cannot give you: memory, storage, and compute
- Reproducible reports with Quarto
- Building a personal website on a web hosting service
- Showcasing projects and writing about your work
- The AI-assisted research workflow

Deliverable. This block ends with a capability check: you demonstrate that your research environment is operational end-to-end and ready to produce work. You will show that:

- **The machine runs.** Your Raspberry Pi is provisioned with the course software stack installed and working.
- **You can reach it.** SSH access from off the network, so the machine is usable from wherever you are working.
- **Code is under version control.** A GitHub repository with your work committed and pushed to a remote.
- **Data comes in.** A successful pull from an API or a database, not a file someone handed you.
- **Work renders reproducibly.** A Quarto document that renders from source to a published output.
- **Work is published.** A live personal website at a public URL, ready to host everything you produce this semester.
- **The AI workflow is in place.** AI tooling configured and integrated into how you work, with a short account of how you are using it.
- **You know your limits.** A brief written statement of what your machine and your host can and cannot give you: memory, storage, compute, and hosting constraints.

Endstate: a working machine you can reach from anywhere, code under version control, and a live personal website that you publish to for the rest of the semester.

5.2 Block 2: Guided Project

A first full pass through the research cycle, worked together as a class on a common dataset.

- Get and clean the data
- Explore and model
- Iterate and refine
- Build and deploy a dashboard
- Present and critique

Endstate: a deployed dashboard and a one page summary, built alongside the class, so you have seen every step of the cycle once before you run it yourself.

5.3 Block 3: Semi-Independent Project

The same cycle again, this time on a question and dataset of your choosing, with the instructor as a coach rather than a guide. We work to checkpoints, and you get a look over your shoulder at every step.

- Scoping your own question
- Acquisition, cleaning, and exploration
- Modeling and iteration
- Build and deploy
- Present and critique

Endstate: a hosted project on a question you chose and defended, run mostly on your own, posted to your website with its one page summary.

5.4 Block 4: Independent Project

The cycle again with the scaffolding off. No checkpoints, no coach at your elbow: you set the schedule and you own the result. This project also has to stretch you, so it must use at least one method or data source you have not worked with before. Open-source intelligence and geospatial analysis are two directions the class may take here.

- Scoping and defending your own question
- Working a method or source that is new to you
- Acquisition, analysis, and iteration on your own timeline
- Build and deploy
- Present and critique

Endstate: a hosted project you ran end to end without a guide, posted to your website with its one page summary.

5.5 Block 5: Capstone Project

The capstone. A real problem, worked through the full cycle, with a public brief at the end. Scoping starts at the midpoint of the semester and runs alongside Block 4, so the capstone has the whole back half of the course to develop rather than the last three weeks. There will be opportunities to work a project for a sponsor.

- Scoping the question
- Data acquisition and analysis
- Mid-point brief
- Build, iterate, and course correct
- Final polish, hosting, and hand-off
- Demo Day: poster and presentations to invited guests

Endstate: a finished, hosted product, a one page summary, and a poster and brief you can present again in the spring.

6 Capstone Project

The capstone is the culminating project of the course: a scoped question, real data, an honest analysis, a hosted deliverable, a one-page written summary, and a Demo Day brief.

There will be opportunities to work a project for a sponsor, most often a faculty member or a Department within USMA, and where the opportunity arises an organization outside the Academy. A sponsor is a good way to land on a problem someone actually cares about, and cadets who want one are encouraged to start the conversation early. LTC Turner will help you find a project either way. No one ends up without one.

Demo Day includes a poster that features your project end-to-end: question, approach, findings, and deliverable. Build it to a Projects Day / MaRS standard.

7 Written Summaries & Portfolio

Every major project in this course (the Guided Project, the Semi-Independent Project, the Independent Project, and the Capstone Project) concludes with a one-page written summary: the question, what you did, what you found, and why it matters, written for an intelligent non-specialist. These one-pagers are hosted on your personal website and collected into a single final portfolio, so by the end of the semester you have a curated, public body of written and analytical work to show advisors, future employers, and program partners.

The writing standard is professional and concise. A one-pager is not a lab notebook dump; it is a deliberate, edited account of a piece of real work. Taken together, these summaries are the written through-line of the course and the written component of your capstone.

8 Assessment & Grading

This is a 1000-point course. Points are distributed across the major blocks as follows:

Block	Points
Setup & Foundations	100
Guided Project	150
Semi-Independent Project	150
Independent Project	200
Capstone Project (incl. Demo Day)	300
Portfolio	100
Total	**1000**

9 AI Policy

AI tools (Claude, GitHub Copilot, ChatGPT, and others) are encouraged in this course. They are not a shortcut; they are the way modern builders work. Learning *how* to work effectively with AI is itself a course objective.

That said, using AI well is not the same as outsourcing your thinking. The standard is simple: you are the author of your work, and you must answer for every part of it.

- **Own every line.** If you cannot explain why a line of code is there, what it does, and why it is correct, it should not be in your submission. AI can draft code quickly, but speed without understanding is a liability, not an asset.
- **Test relentlessly.** AI-generated code can look right and be wrong. It can hallucinate function names, invert logic, or silently drop edge cases. You are responsible for verifying that your code runs correctly and produces correct results. "The AI wrote it" is not a defense.

- **You present, you defend.** When you stand up to present your project or submit your work, it is *yours*. You will be asked questions. You will need to justify your choices. If you cannot, it does not matter how polished the deliverable looks.
- **Use AI to learn, not to skip learning.** The best use of AI in this course is as a thought partner: ask it to explain concepts, debug errors with you, suggest approaches you had not considered. The worst use is pasting a prompt and submitting the output without engaging your own judgment.

There are no restrictions on *when* or *how often* you use AI. The restriction is on presenting work you do not understand as your own. That is true with or without AI. It is simply the standard we hold as professionals.

10 Required Materials

- **Raspberry Pi** (provided on loan for the semester)
- **Laptop** with SSH client and web browser
- **GitHub account** (free)
- **Web hosting account** (free tier, service announced in class)
- **Tailscale account** (free tier)